

Our Digital World — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper. Every sum is built to work by hand, using small numbers or powers of ten.
- A label alone is rarely a full answer: where a question says **explain** or **describe**, give the action or the reason. A defence only scores with its reason — “install a firewall” earns nothing; say what it does that stops the attack.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe situations you may not have seen before. You are not expected to recognise them — apply the ideas you already have.

SECTION 1 – RECALL

1. Write a definition of each term, in one sentence.

(a) Active digital footprint

[1]

(b) Passive digital footprint

[1]

(c) Misinformation

[1]

(d) Copyright

[1]

2. Before you believe or share a post, you run five checks. Name **three** of them, and for each say what you would actually look at. [3]

3. Catfishing, trolling, fraping and dissing are four different acts. For each word, say what the person **does**, in ten words or fewer, without using the word itself.

(a) Catfishing [1]

(b) Trolling [1]

(c) Fraping [1]

(d) Dissing [1]

SECTION 2 – WORKING FROM PRINTED MATERIAL

4. Farhan's photograph is on a school-resources website. Four students want to use it. For each one, state whether the licence allows it, and name the condition in the licence that decides it. A mark needs both.

LICENCE NOTICE · SCHOOL-RESOURCES WEBSITE	
TITLE	Mangoes in the Monsoon
CREATOR	Farhan Qureshi
SOURCE	openphotos.example/farhan-q
LICENCE	Creative Commons BY-NC-SA

(a) Ria puts it in a free article on her school's website, with a credit line. [1]

- (b) Ravi crops the photo, adds the words “Rain, rain, go away”, credits Farhan, and shares the poster free online. State what the licence lets him do **and** what he must do with the poster. [2]

- (c) Nisha prints it on the brochure for her paid tuition classes, with a credit line. [1]

- (d) Kunal copies it into his video with no credit, saying “everybody does it”. [1]

5. A local website runs this headline. The police figures are printed beneath it.

LOCAL NEWS SITE • WARD 9 DAILY	
HEADLINE	CHAIN-SNATCHING IN WARD 9 UP 150% THIS MONTH!
POLICE FIGURES	Last month: 4 cases · This month: 10 cases

- (a) Calculate the percentage increase. [1]

- (b) Next month there are 4 cases again. A friend says: “So it fell by 150%.” Calculate the real percentage fall. [1]

- (c) The headline is arithmetically correct. State one thing it hides. [1]

6. A password is made of **4 random lowercase letters**. An attacker's program makes 1,000 guesses a second. You are given $26 \times 26 = 676$. You have no calculator, so use estimates that you can check by hand.
- (a) Show that there are more than 100,000 possible passwords. (A 5-digit PIN has exactly 100,000.) [1]
-
-
-
- (b) Show that there are fewer than 1,000,000 possible passwords. (A 6-digit PIN has exactly 1,000,000.) [1]
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-
-
- (c) Between which two times does the attacker take to try them all? Give both in seconds. [1]
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SECTION 3 – SITUATIONS YOU MAY NOT HAVE SEEN

7. Nandini runs a boutique in Hubballi. Her firewall is on and her antivirus is updated every night. An assistant installs a free “billing theme” app on the shop laptop. Over the next fortnight the app secretly records everything typed, including the shop's net-banking password, and sends it to a stranger, who then logs in to the account from another city. Nobody notices anything wrong with the laptop.
- (a) Name the malware. [1]
-
- (b) Name the guard Nandini did not have, and state what it does. [1]
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- (c) Explain why the firewall did not stop this. [1]
-
-
-

(d) State one habit that would have prevented the infection in the first place. [1]

8. Tanvi’s friend tells her what happened over nine weeks. Read the timeline.

WHAT HAPPENED · IN ORDER	
WEEK 1	A new account, “Riya_13”, follows Tanvi. Its photos show a girl in a school uniform.
WEEKS 2–8	They chat every night. Tanvi shares secrets, including that she is scared of speaking on the school stage.
WEEK 9	A reverse image search shows the photos belong to a model’s public page in another country.
WEEK 9	“Riya_13” posts Tanvi’s secret in a group chat of 40 classmates, with the caption “Our stage star, ha ha ha”.

(a) Two of the four named acts happened here. Name both, and say which part of the story shows each. [2]

(b) Tanvi wants to delete the whole chat and her account straight away. Explain why she should save evidence first. [1]

(c) Tanvi’s friend wants to send “Riya_13” an angry threat. Explain why that is a bad idea. [1]

End of paper. Check that every defence you named comes with its reason, and every calculation with its working.